

PROCÉDURE OFFICIELLE DE CONFIGURATION

HSRP & EtherChannel

Infrastructure Réseau Redondante

Document pédagogique – Usage académique

Miri Mohammed et Fayol Nkalou

A. Réinitialisation des équipements

```
Commutateurs :  
Switch# erase startup-config  
Switch# delete flash:vlan.dat  
Switch# reload
```

```
Routeur :  
Router# erase startup-config  
Router# reload
```

B. Nommage des équipements

```
Switch(config)# hostname SW1  
Switch(config)# hostname SW2  
Router(config)# hostname R1
```

C. Création des ports Trunk (SW1)

```
SW1(config)# interface fastEthernet 0/22  
SW1(config-if)# switchport mode trunk
```

D. Création des VLANs (SW1)

```
SW1(config)# vlan 10  
name Administration  
exit SW1(config)#  
vlan 20 name Equipes  
exit SW1(config)#  
vlan 30 name WiFi
```

```
SW1(config)#vlan 10  
SW1(config-vlan)#administration  
^  
% Invalid input detected at '^' marker.  
  
SW1(config-vlan)#name administration  
SW1(config-vlan)#exit  
SW1(config)#vlan 20  
SW1(config-vlan)#name equipes  
SW1(config-vlan)#exit  
SW1(config)#vlan 30  
SW1(config-vlan)#name wifi  
SW1(config-vlan)#exit  
SW1(config)#  
SW1(config)#
```

E. Attribution des ports aux VLANs (SW1)

```
interface range fastEthernet 0/1 - 6  
switchport access vlan 10
```

```
interface range fastEthernet 0/7 - 12  
switchport access vlan 20
```

```
interface range fastEthernet 0/13 - 14  
switchport access vlan 30
```

F. Création des VLANs et ports (SW2)

```
SW2(config)# vlan 10
name Administration
SW2(config)# vlan 20
name Equipes
SW2(config)# vlan 30
name WiFi

interface range fastEthernet 0/1 - 6
switchport access vlan 10

interface range fastEthernet 0/7 - 12
switchport access vlan 20

interface range fastEthernet 0/13 - 14
switchport access vlan 30
```

```
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#
Switch(config)#hostname SW2
SW2(config)#vlan 10
SW2(config-vlan)#name administration
SW2(config-vlan)#vlan 20
SW2(config-vlan)#equipes
      ^
% Invalid input detected at '^' marker.

SW2(config-vlan)#name equipes
SW2(config-vlan)#vlan 30
SW2(config-vlan)#name wifi
SW2(config-vlan)#exit
SW2(config)#interface range fastEthernet 0/1 - 6
SW2(config-if-range)#switchport access vlan 10
SW2(config-if-range)#interface range fastEthernet 0/7 - 12
SW2(config-if-range)#switchport access vlan 20
SW2(config-if-range)#interface range fastEthernet 0/13 - 14
SW2(config-if-range)#switchport access vlan 30
SW2(config-if-range)#exit
SW2(config)#exit
```

G. Routage Inter-VLAN (R1)

```
interface fastEthernet 0/0
no shutdown

interface fastEthernet 0/0.10
encapsulation dot1Q 10 ip address
172.20.0.1 255.255.255.0

interface fastEthernet 0/0.20
encapsulation dot1Q 20 ip address
172.20.1.1 255.255.255.0

interface fastEthernet 0/0.30
encapsulation dot1Q 30 ip address
172.20.2.1 255.255.255.128
```

```

Router(config-subif)#interface fa0/0.10
Router(config-subif)#encapsulation dot1q 10
Router(config-subif)#ip address 172.20.0.2 255.255.255.0
Router(config-subif)#exit
Router(config)#interface fa0/0.20
Router(config-subif)#encapsulation dot1q 20
Router(config-subif)#ip address 172.20.1.2 255.255.255.0
Router(config-subif)#exit
Router(config)#interface fa0/0.30
Router(config-subif)#encapsulation dot1q 30
Router(config-subif)#ip address 172.20.2.2 255.255.255.128
Router(config-subif)#exit
Router(config)#interface fa0/1
Router(config-if)#no shutdown
Router(config-if)#
*Jan 1 02:29:32.487: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
Router(config-if)#exit

```

H. NAT / PAT (R1)

```

interface fastEthernet 0/1
ip address dhcp ip nat
outside

```

```

interface fastEthernet 0/0.10
ip nat inside interface
fastEthernet 0/0.20 ip nat
inside interface fastEthernet
0/0.30 ip nat inside

```

```

access-list 10 permit 172.20.0.0 0.0.0.255
access-list 20 permit 172.20.1.0 0.0.0.255
access-list 30 permit 172.20.2.0 0.0.0.127

```

```

r1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
r1(config)#interface fa0/1
r1(config-if)#ip address dhcp
r1(config-if)#ip nat outside

*Jan 24 14:58:57.403: %LINEPROTO-5-UPDOWN: Line protocol on Interface NVI0, changed state to up
r1(config-if)#
r1(config-if)#interface fastEthernet 0/0.10
r1(config-subif)#ip nat inside
r1(config-subif)#exit
r1(config)#interd
r1(config)#interface fa0/0.20
r1(config-subif)#ip nat inside
r1(config-subif)#exit
r1(config)#interface fa0/0.30
r1(config-subif)#ip nat inside
r1(config-subif)#exit
r1(config)#access-list 10 permit 172.20.0.0 0.0.0.255
r1(config)#access-list 20 permit 172.20.1.0 0.0.0.255
r1(config)#access-list 30 permit 172.20.2.0 0.0.0.127
^

```

```

ip nat inside source list 10 interface fastEthernet 0/1 overload
ip nat inside source list 20 interface fastEthernet 0/1 overload
ip nat inside source list 30 interface fastEthernet 0/1 overload

```

```

ip route 0.0.0.0 0.0.0.0 10.0.228.1

```

```

r1(config)#access-list 30 permit 172.20.2.0 0.0.0.127
r1(config)#ip nat inside source list 10 interface fa0/1 overload

r1(config)#ip nat inside source list 20 interface fa0/1 overload

r1(config)#ip nat inside source list 30 interface fa0/1 overload

r1(config)#exit

```

I. Routage Inter-VLAN & NAT (R2)

```
interface fastEthernet 0/0
no shutdown

interface fastEthernet 0/0.10
encapsulation dot1Q 10 ip address
172.20.0.2 255.255.255.0

interface fastEthernet 0/0.20
encapsulation dot1Q 20 ip address
172.20.1.2 255.255.255.0

interface fastEthernet 0/0.30
encapsulation dot1Q 30 ip address
172.20.2.2 255.255.255.128 (NAT
identique à R1 + route par défaut)
```

J. Configuration HSRP

```
R1 (actif) :
interface FastEthernet0/0.10
standby 10 ip 172.20.0.3
standby 10 priority 150
standby 10 preempt

interface FastEthernet0/0.20

standby 20 ip 172.20.1.3
standby 20 priority 150
standby 20 preempt

interface FastEthernet0/0.30
standby 30 ip 172.20.2.3
standby 30 priority 150
standby 30 preempt
```

```
r1>enable
r1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
r1(config)#interface fa0/1
r1(config-if)#no shutdown
r1(config-if)#
*Jan 24 15:41:02.491: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
r1(config-if)#exit
r1(config)#interface fa0/0.10
r1(config-subif)#standby 10 ip 172.20.0.3
r1(config-subif)#standby 10 priority 150
r1(config-subif)#standby 10 preempt
*Jan 24 15:41:59.067: %HSRP-5-STATECHANGE: FastEthernet0/0.10 Grp 10 state Standby -> Active
r1(config-subif)#exit
r1(config)#interface fa0/0.20
r1(config-subif)#standby 20 ip 172.20.1.3
r1(config-subif)#standby 20 priority 150
r1(config-subif)#standby 20 preempt
r1(config-subif)#exit
r1(config)#interface fa
*Jan 24 15:42:54.175: %HSRP-5-STATECHANGE: FastEthernet0/0.20 Grp 20 state Standby -> Active0
% Incomplete command.

r1(config)#interface fa0/0.30
r1(config-subif)#standby 30 ip 172.20.2.3
r1(config-subif)#standby 30 priority 150
r1(config-subif)#standby 30 preempt
r1(config-subif)#
*Jan 24 15:43:52.943: %HSRP-5-STATECHANGE: FastEthernet0/0.30 Grp 30 state Standby -> Active
r1(config-subif)#
```

```
R2 (secours) :
interface FastEthernet0/0.10
standby 10 ip 172.20.0.3
standby 10 priority 90
```

```

interface FastEthernet0/0.20
standby 20 ip 172.20.1.3
standby 20 priority 90

interface FastEthernet0/0.30
standby 30 ip 172.20.2.3

standby 30 priority 90

```

K. EtherChannel & Trunks (SW1 / SW2)

Trunk :

```

interface fastEthernet 0/22
switchport mode trunk

```

EtherChannel :

```

interface range fastEthernet 0/23 - 24
channel-group 1 mode desirable

```

Trunk sur Port-Channel :

```

interface port-channel 1
switchport mode trunk

```

```

SW1>enable
SW1#config t
Enter configuration commands, one per line. End with CNTL/Z.
SW1(config)#interface fa0/22
SW1(config-if)#switchport mode trunk
SW1(config-if)#interface range
^
% Invalid input detected at '^' marker.

SW1(config-if)#exit
SW1(config)#interface range fa0/23 - 24
SW1(config-if-range)#channel-group 1 mode desirable
Creating a port-channel interface Port-channel 1

SW1(config-if-range)#
*Mar 1 01:59:38.282: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to down
*Mar 1 01:59:40.270: %LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed state to up
*Mar 1 01:59:41.050: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up
*Mar 1 01:59:42.032: %LINK-3-UPDOWN: Interface Port-channel1, changed state to up
SW1(config-if-range)#switchport mode trunk
Command rejected: An interface whose trunk encapsulation is "Auto" can not be configured to "trunk" mode.
% Range command terminated because it failed on FastEthernet0/23
SW1(config-if-range)#switchport trunk encapsulation dot1q
SW1(config-if-range)#
*Mar 1 02:00:50.768: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to down
*Mar 1 02:00:50.768: %LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed state to down
*Mar 1 02:00:51.783: %LINK-3-UPDOWN: Interface Port-channel1, changed state to down
*Mar 1 02:00:53.595: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up
*Mar 1 02:00:54.577: %LINK-3-UPDOWN: Interface Port-channel1, changed state to up
*Mar 1 02:00:55.583: %LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed state to up
SW1(config-if-range)#switchport mode trunk
SW1(config-if-range)#

```